Task 2 – Basic operation

Algorithm	Basic operation	Why it dominates
Linear Search	Equality test A[i] == target	The scan repeats key comparisons until a match is found or the end is reached. The count grows with the position and input size \(n\).
Recursive Binary Search	Compare target with the middle key	Each comparison discards about half the remaining interval.
Hash Table	Hash computation and bucket/key lookup	Expected one indexed lookup; collision resolution may require extra key comparisons.